Intelligent Telecommunication Automation System

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Ankit Yadav - EN22CS301136
Anshuman Geete - EN22CS301160
Arti Jain - EN22CS301206
Atisha Jain - EN22CS301235
Avani Sharma - EN22CS301239
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Under the Guidance of

Prof.(Dr.) Hemlata Patel
Prof. Ajeet Singh Rajput
Prof. Akshay Saxena

MEDICAPS
UNIVERSITY

Department of Computer Science & Engineering
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MEDICAPS UNIVERSITY, INDORE- 453331

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JAN-JUNE 2026

Report Approval

The project work **"Intelligent Telecommunication Automation System"** is hereby approved as a creditable study of an engineering subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.
It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation:

Affiliation

External Examiner

Name:

Designation

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Declaration

We hereby declare that the project entitled **"Intelligent Telecommunication Automation System"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology of Computer Science in 'B-TECH' completed under the supervision of **Prof. (Dr.)Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena ,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

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Certificate

We, **Prof.(Dr.)Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena** certify that the project entitled **" Intelligent Telecommunication Automation system "** submitted in partial fulfillment for the award of the degree of Bachelor of Technology of Computer Science by **Ankit Yadav, Anshuman Geete, Arti Jain, Atisha Jain, Avani Sharma, Ishana Hatodiya,** is the record carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Prof. Ajeet Singh Rajput	Mr. Ashwin Krishnan
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Dr. Kailash Chandra Bandhu

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I express my heartfelt gratitude to my **External Guides -** Mr. Suraj Naik, Mr. Ashwin Krishnan, Mr. Vaibhav as well as to my **Internal Guides** - Prof.(Dr)Hemlata Patel, Prof. Ajeet Singh Rajput, Prof. Akshay Saxena without whose continuous help and support, this project would ever have reached to the completion.
I would also like to thank to my team who extended their kind support and help towards the completion of this project.

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Without their support this report would not have been possible.

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Abstract

The rapid growth of telecom services has increased the need for efficient customer support systems. Traditional methods rely on manual processes, leading to delays and reduced user satisfaction. This project presents an Intelligent Telecommunication Automation System that integrates Generative AI and Agentic AI with automated DevOps practices using Infrastructure as Code (IaC).

The system is a chatbot-based platform designed to handle telecom-related queries such as network issues, data problems, recharge failures, and service configurations. It uses AI-driven response generation to understand user queries and provide accurate solutions. Additionally, agentic capabilities enable intelligent actions like retrieving subscriber data and suggesting troubleshooting steps.

On the DevOps side, infrastructure provisioning is automated using Terraform, with Ansible for configuration management. Docker ensures consistent environments through containerization, while GitHub Actions enables CI/CD on AWS cloud infrastructure.

The system supports both local deployment using Docker Compose and cloud deployment via AWS EC2, ensuring scalability and flexibility. Monitoring tools provide insights into system performance and health.

Overall, this project demonstrates the integration of AI-driven solutions with DevOps automation to build a scalable and efficient telecom support system.

Keywords: *Agentic AI ,Generative AI ,DevOps ,Infrastructure as Code ,Terraform ,Ansible ,Docker ,AWS ,CI/CD ,Telecom Chatbot ,Automation*

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Abbreviations

AI	Artificial Intelligence
IaC	Infrastructure as Code
CI/CD	Continuous Integration / Continuous Deployment
AWS	Amazon Web Services
VPC	Virtual Private Cloud
EC2	Elastic Compute Cloud
NLP	Natural Language Processing

Notations & Symbols

API	Application Programming Interface
IP	Internet Protocol
HTTP	Hypertext Transfer Protocol

CHAPTER-1

INTRODUCTION

1.1 Introduction

The telecommunications industry plays a crucial role in connecting people and enabling communication across the globe. With the rapid increase in mobile users and internet-based services, telecom providers are required to manage an enormous volume of customer interactions on a daily basis. These interactions typically involve issues such as network connectivity problems, billing discrepancies, recharge failures, slow internet speeds, and service activation or deactivation.

Traditional telecom support systems rely heavily on human customer service representatives. While these systems are capable of handling complex queries, they suffer from several limitations such as long waiting times, high operational costs, inconsistent service quality, and lack of scalability during peak hours. As the number of users continues to grow, these limitations become more significant and impact overall customer satisfaction.

In recent years, Artificial Intelligence (AI) has emerged as a powerful tool for automating customer support systems. In particular, Generative AI has enabled the development of intelligent chatbots capable of understanding natural language, interpreting user intent, and generating human-like responses. However, most existing AI chatbots are limited to providing responses and do not possess the ability to perform real actions or interact deeply with backend systems.

This project introduces an Agentic AI-Based Intelligent Telecom Support System, which goes beyond traditional chatbot functionality. The system not only understands user queries but also takes intelligent actions such as retrieving subscriber data, diagnosing issues, and suggesting solutions. This is achieved through the integration of agentic behavior, where the system can make decisions and execute tasks autonomously.

In addition to AI capabilities, the project incorporates modern DevOps practices to automate infrastructure provisioning and deployment. Using Infrastructure as Code (IaC), tools such as Terraform and Docker ensure that the system can be deployed consistently across different

environments. Continuous Integration and Continuous Deployment (CI/CD) pipelines using GitHub Actions further automate the development and deployment process.

The combination of AI and DevOps results in a system that is scalable, efficient, reliable, and suitable for real-world telecom applications

.

1.2 Literature Review

Telecom customer support systems have evolved significantly over the years. Initially, support systems were entirely manual, relying on human agents to resolve customer issues. While effective, these systems were time-consuming and expensive.

Rule-based chatbots were introduced as an early form of automation. These systems operated on predefined rules and decision trees, allowing them to respond to simple queries. However, they lacked flexibility and could not handle complex or unexpected inputs.

With advancements in Natural Language Processing (NLP), AI-based chatbots were developed to improve interaction quality. These systems could understand user queries to some extent but were still limited in their ability to perform actions or integrate with backend systems.

Recent developments in Generative AI have enabled chatbots to generate dynamic and context-aware responses. However, most implementations still lack agentic capabilities, meaning they cannot independently decide actions or interact with external systems.

In parallel, DevOps practices have transformed the way software systems are developed and deployed. Infrastructure as Code (IaC) allows developers to define infrastructure using code, reducing manual errors and improving reproducibility. Tools like Docker enable containerization, ensuring consistent environments across development and production. CI/CD pipelines automate testing and deployment, increasing efficiency and reliability.

Despite these advancements, there is still a gap in integrating AI-based support systems with automated infrastructure. This project aims to bridge this gap by combining Generative AI, Agentic AI, and DevOps practices into a unified system.

1.3 Problem Statement

Telecom companies face several challenges in providing efficient customer support. These include handling a large volume of queries, maintaining consistent service quality, reducing response time, and managing infrastructure efficiently.

Existing systems are either manual, which are slow and costly, or partially automated, which lack intelligence and scalability. There is a need for a system that can provide intelligent, real-time support while also ensuring efficient deployment and scalability

1.4 Objectives

- Design and develop an intelligent chatbot for telecom support
- Implement Generative AI for contextual response generation
- Integrate Agentic AI for decision-making and task execution
- Automate infrastructure provisioning using Terraform
- Implement configuration management using Ansible
- Containerize services using Docker
- Build an automated CI/CD pipeline using GitHub Actions
- Deploy the system on AWS cloud
- Ensure scalability, reliability, and fault tolerance

1.5 Significance of the Study

- Enhances customer experience through instant support
- Reduces dependency on human agents
- Demonstrates real-world integration of AI and DevOps
- Provides a scalable and cost-effective solution for telecom companies
- Bridges the gap between intelligent applications and automated infrastructure

1.6 Research Design

The project follows a systematic approach:

- Problem Identification → Inefficient telecom support
- Solution Design → AI chatbot + DevOps pipeline
- Implementation → AI + Cloud + IaC
- Testing → Functional and deployment testing
- Evaluation → Performance and scalability

1.7 Source of Data

- User input queries
- Telecom simulation services
- AI-generated outputs
- System logs and monitoring data

1.8 Chapter Scheme

- Chapter 1: Introduction
- Chapter 2: System Design and Architecture
- Chapter 3: Implementation Details
- Chapter 4: Results and Discussion
- Chapter 5: Conclusion
- Chapter 6: Future Scope

CHAPTER 2

SYSTEM DESIGN & METHODOLOGY

2.1 System Architecture

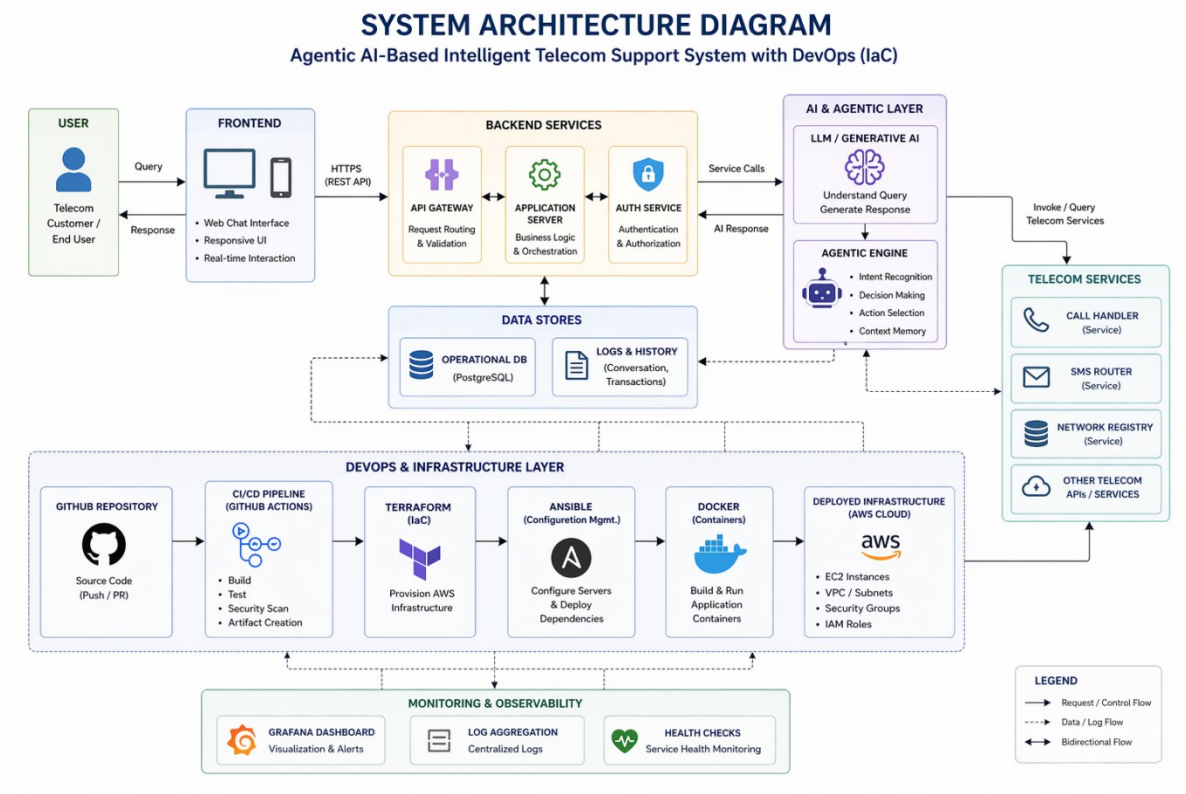


Fig 2.1.1

The system architecture diagram illustrates the overall structure of the telecom support system. It shows how the user interacts with the frontend interface, which communicates with the backend server. The backend integrates with the AI module and agentic layer to process queries and generate responses. The DevOps layer automates deployment, while the infrastructure layer provides cloud resources.

The system architecture is designed to integrate Artificial Intelligence with DevOps practices. It consists of multiple layers that work together to deliver the final output.

Architecture Layers

1. User Layer

This is the front-end interface where users interact with the system. It allows users to:

- Enter queries
- View responses
- Interact with chatbot

2. Application Layer

This layer acts as a bridge between the user interface and backend services. It:

- Handles API requests
- Processes input data
- Sends responses

3. AI Layer

The AI layer is responsible for:

- Understanding user queries
- Identifying intent
- Generating responses

4. Agentic Layer

This layer adds intelligence to the system by:

- Deciding actions
- Executing backend calls
- Performing operations

5. DevOps Layer

This layer ensures:

- Automated deployment
- Continuous integration
- Infrastructure management

6. Infrastructure Layer

This includes:

- AWS EC2 instances
- Networking (VPC)
- Security groups

2.2 USE CASE DIAGRAM

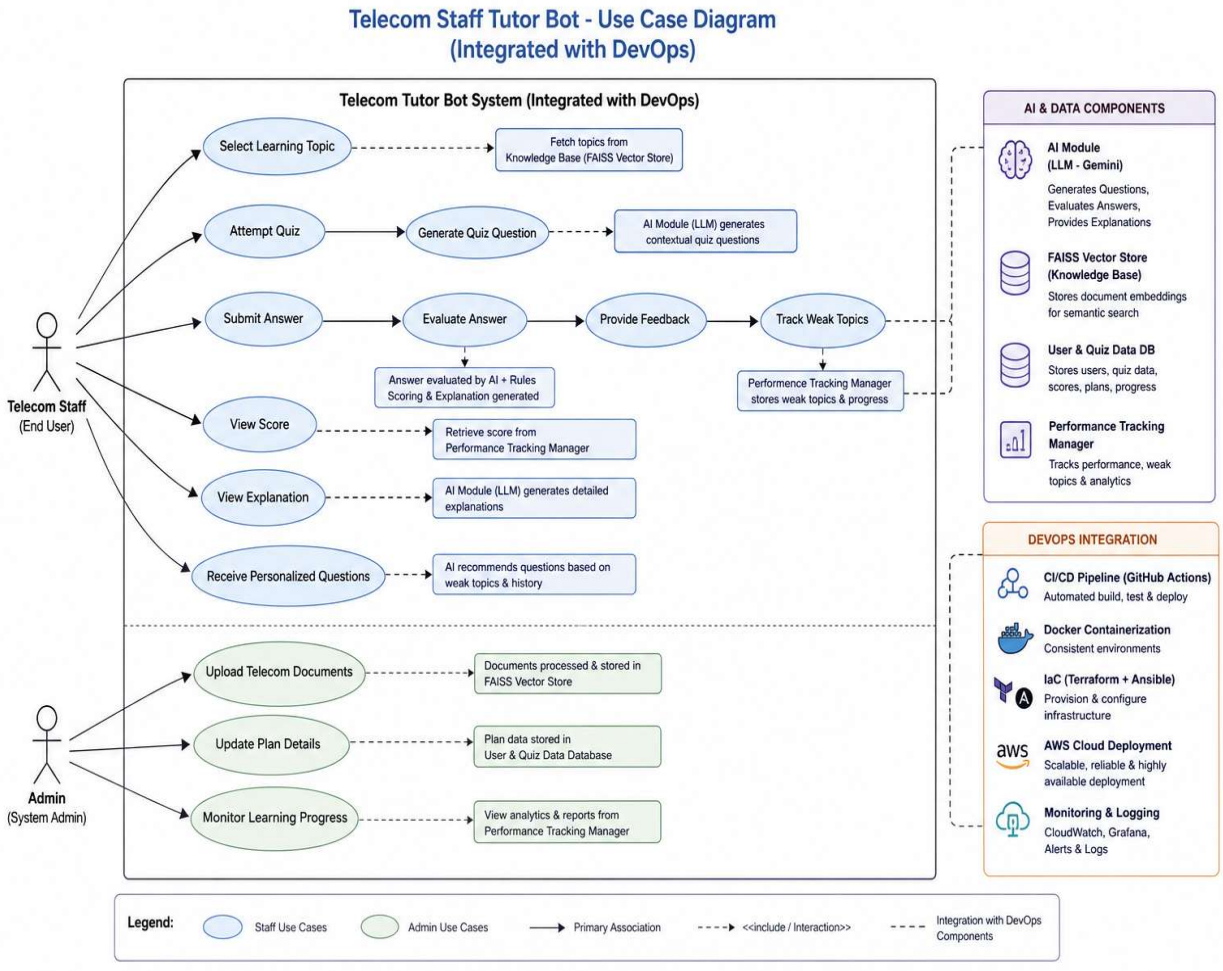


Fig 2.2.1

2.3 Data Flow Diagram (DFD)

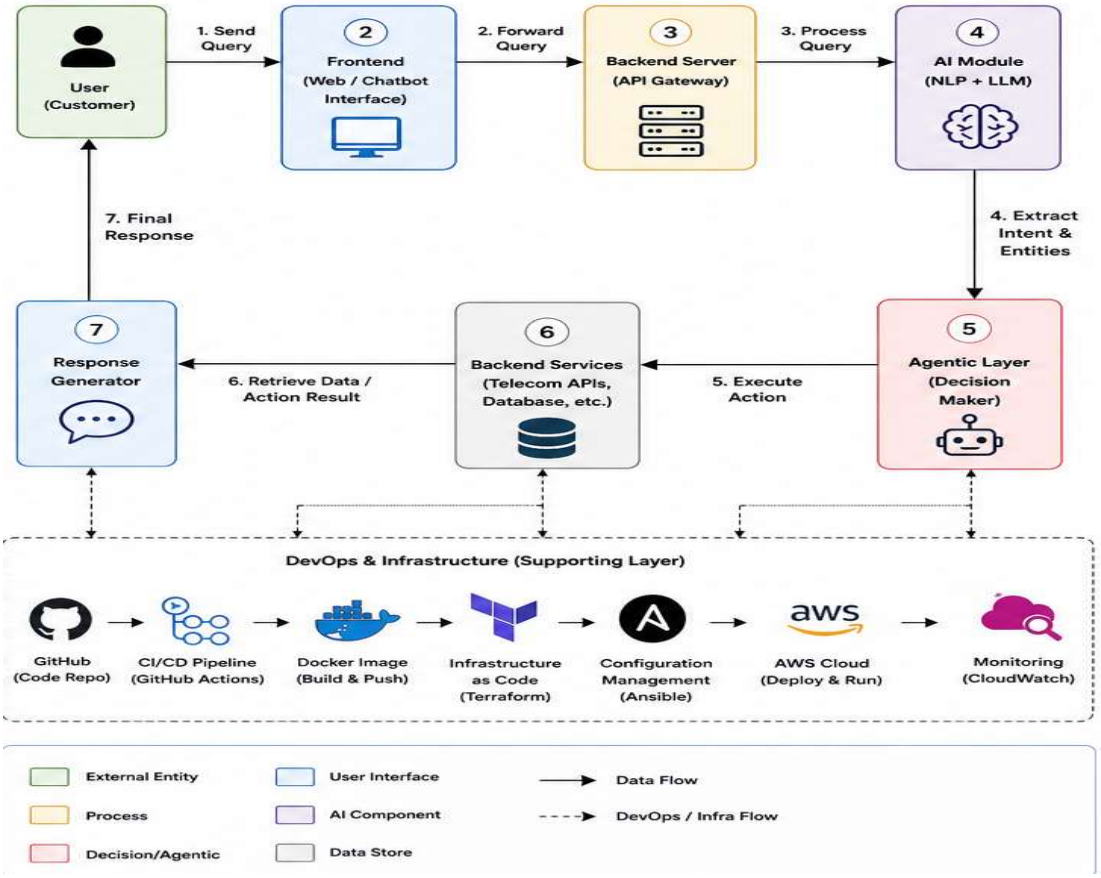


Fig 2.3.1

The Data Flow Diagram shows how data moves through the system. It starts from user input, passes through backend processing, AI interpretation, and agentic decision-making, and finally returns a response to the user. This diagram helps in understanding system interactions and data handling.

Data Flow Steps

1. User sends query
2. Request reaches backend
3. AI processes input
4. Agent decides action
5. Backend retrieves data
6. Response sent to user

This ensures smooth communication between system components.

2.4 Sequence Diagram

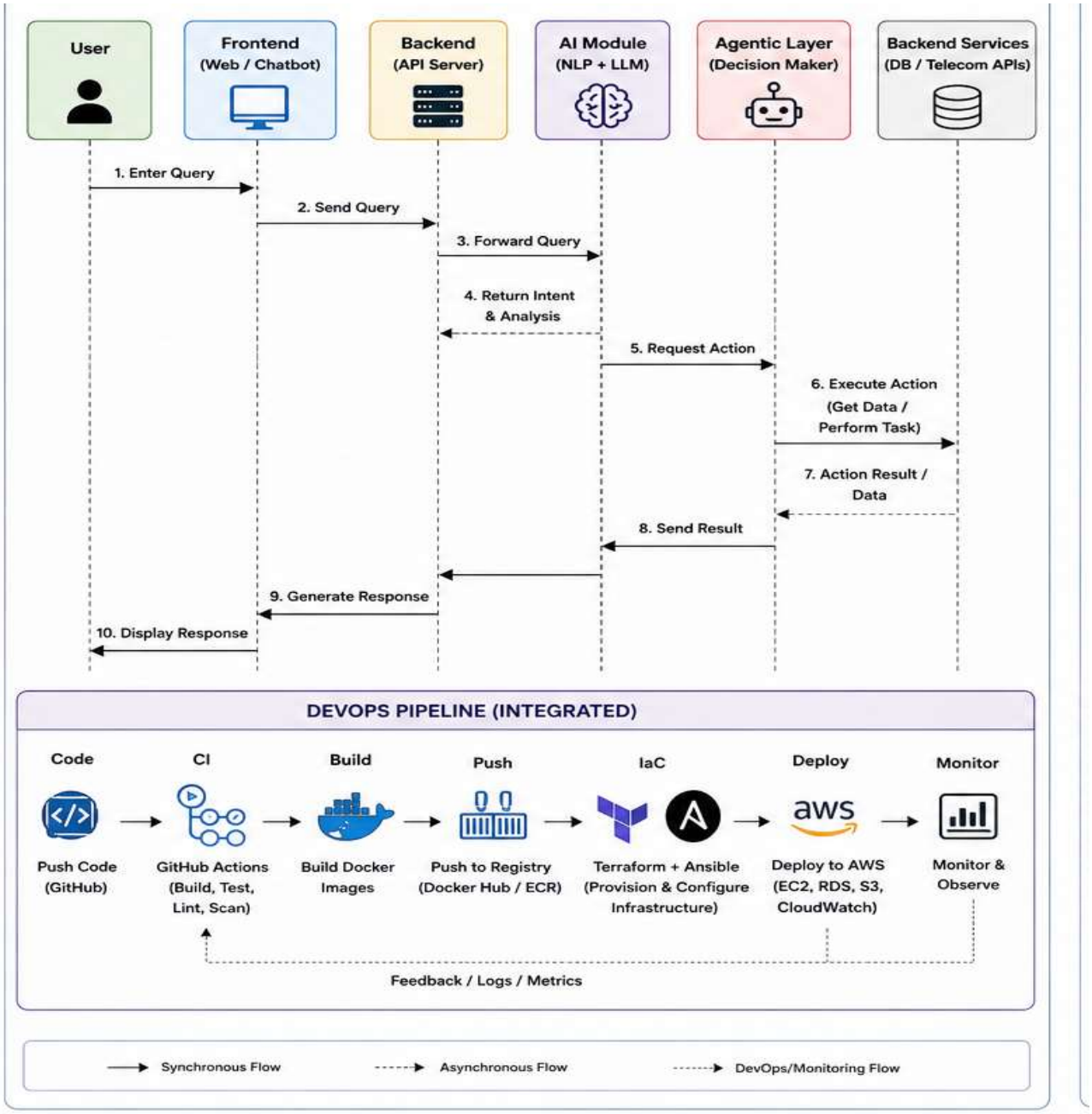


Fig 2.4.1

The sequence diagram represents the order of interactions between system components. It shows how the user query flows through frontend, backend, AI module, and agent before returning a response. It highlights the time-based execution of the system

2.5 Workflow Diagram

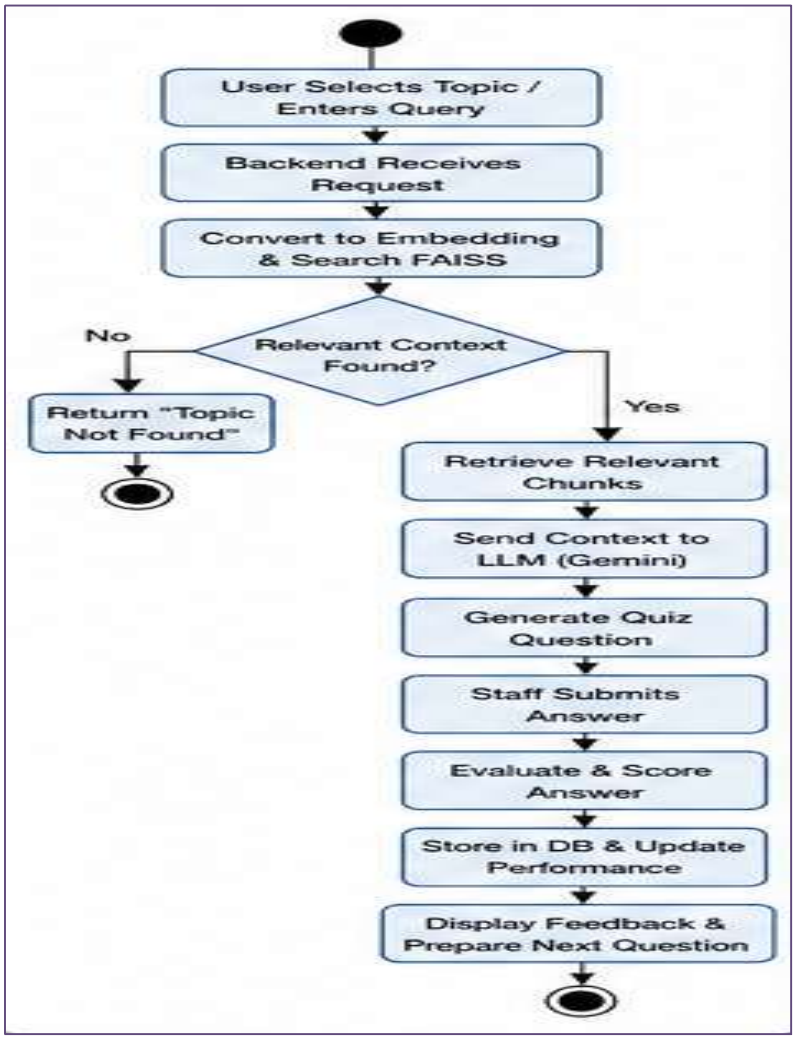


Fig 2.5.1

- User selects topic
- Backend processes request
- Convert to embeddings
- Search FAISS database
- Retrieve relevant content
- Generate quiz via LLM
- User submits answer
- Evaluate and score answer
- Store performance data
- Provide feedback & next question

2.6 Component Diagram

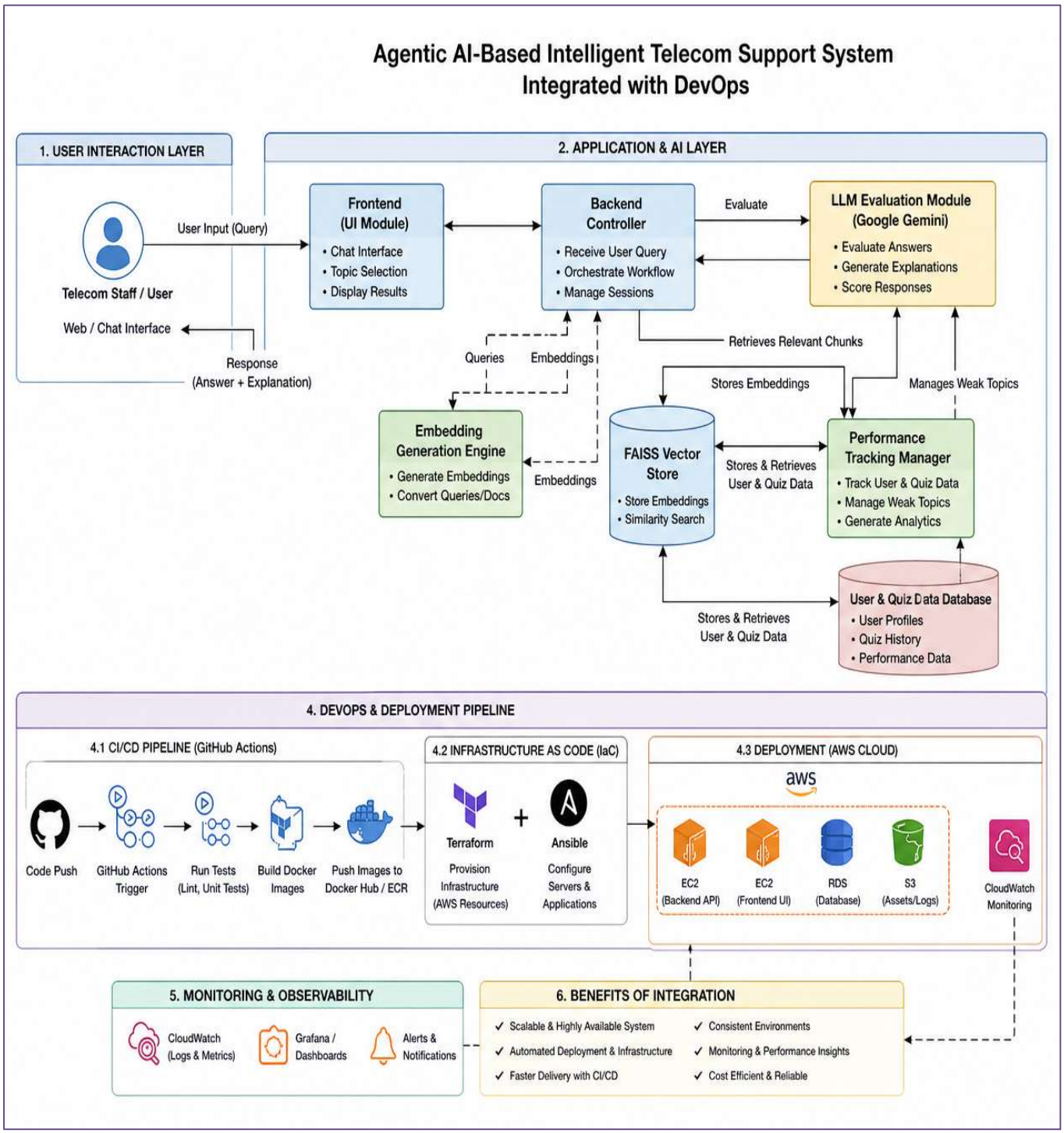


Fig 2.6.1

2.7 Methodology

The system follows a structured methodology:

Step 1: Query Input

User enters query through interface

Step 2: Processing

Backend processes request

Step 3: AI Analysis

AI model interprets query

Step 4: Decision Making

Agent decides action

Step 5: Execution

Backend services are accessed

Step 6: Response Generation

Final output is generated

2.8 System Design Approach

The system is designed using a modular approach:

- Each component works independently
- Easy to update and maintain
- Scalable architecture

2.9 Technologies Used

Frontend

- Web-based interface

Backend

- FastAPI / Flask

AI

- Generative AI model

DevOps

- Terraform
- Docker
- GitHub Actions

Cloud

- AWS EC2

2.10 Infrastructure Design

The infrastructure is designed using Infrastructure as Code (IaC).

Components

- EC2 instance for hosting
- Security groups for access control
- Docker containers for application

2.11 CI/CD Pipeline Design

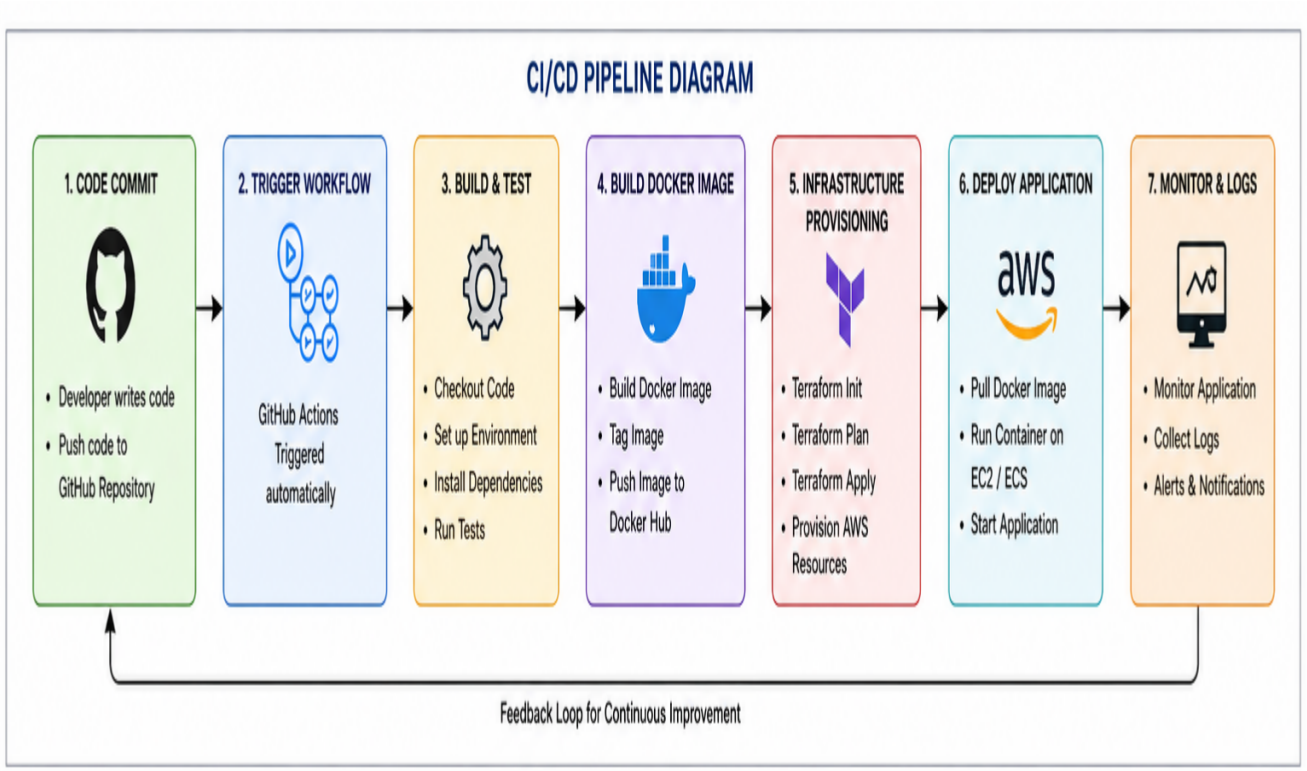


Fig 2.11.1

The CI/CD pipeline diagram illustrates the automated deployment workflow of the system. It begins when code changes are pushed to the GitHub repository. This action triggers GitHub Actions to start the CI/CD pipeline. The pipeline performs automated tasks such as code building, linting, and testing. Once validated, Docker is used to containerize the application for consistency across environments. The pipeline then uses Terraform to provision cloud infrastructure on AWS. Ansible is used to configure servers and deploy application components. Finally, the application is deployed on AWS services, ensuring a scalable, reliable, and fully automated deployment process.

Pipeline Steps

1. Code push
2. Build process
3. Testing
4. Deployment

2.12 Advantages of the Design

- **Scalable:** The system can easily handle increasing numbers of users and requests by leveraging cloud infrastructure and containerized services.
- **Reliable:** Automated deployment and monitoring ensure consistent performance with minimal downtime.
- **Automated:** DevOps practices such as CI/CD pipelines and Infrastructure as Code reduce manual intervention and human errors.
- **Efficient:** AI-powered responses and automation significantly reduce response time and improve overall system performance.
- **Flexible:** Supports both local (Docker Compose) and cloud-based (AWS) deployments, making it adaptable to different environments.

2.13 Limitations of Design

- **Dependent on Cloud Availability:** System performance may be affected if cloud services (e.g., AWS) experience downtime.
- **Requires Internet Connection:** Users must have stable internet access to interact with the system.
- **AI Accuracy Limitations:** AI-generated responses may sometimes be incorrect or less precise, especially for complex queries.
- **Initial Setup Complexity:** Setting up DevOps tools and infrastructure requires technical expertise.
- **Cost Factors:** Cloud infrastructure and continuous deployment pipelines may incur operational costs.
- **Data Dependency:** The system's effectiveness depends on the quality and relevance of training data and stored knowledge.

CHAPTER 3

IMPLEMENTATION

This chapter describes the implementation details of the Agentic AI-Based Intelligent Telecom Support System. It covers the development of the AI chatbot, integration of agentic behavior, backend services, and the DevOps pipeline used for automated deployment.

The implementation phase focuses on converting the designed architecture into a working system by integrating multiple technologies such as Generative AI, Docker, Terraform, and AWS.

3.1 System Development Environment

The system was developed using the following environment:

Hardware Environment

- Processor: Intel i5 or above
- RAM: Minimum 4GB (8GB recommended)
- Storage: 20GB free space

Software Environment

- Operating System: Windows/Linux
- Programming Language: Python
- Framework: FastAPI / Flask (backend)
- Containerization: Docker
- Infrastructure: Terraform
- Cloud Platform: AWS
- Version Control: GitHub

3.2 AI Chatbot Implementation

The chatbot is the core component of the system responsible for interacting with users and providing solutions to telecom-related problems.

Working Mechanism

1. User inputs a query through the interface
2. The query is sent to the backend API
3. The AI model processes the input
4. Response is generated based on context

Key Features

- Natural language understanding
- Context-aware responses
- Multi-turn conversation handling
- Telecom-specific query resolution

The chatbot uses prompt-based techniques to ensure relevant and meaningful responses.

3.3 Agentic AI Functionality

The system incorporates agentic behavior, enabling it to take intelligent actions rather than just responding.

Agent Workflow

1. Receive user query
2. Identify intent
3. Decide action
4. Execute backend operation
5. Generate final response

Example

Query: “My data is not working”

- AI identifies issue → data problem
- Agent checks possible causes
- Suggests troubleshooting steps

Advantages

- Improves problem-solving capability
- Enables real-time actions
- Enhances user experience

3.4 Backend Implementation

The backend handles communication between the user interface, AI module, and telecom services.

Responsibilities

- API request handling
- Data processing
- Integration with AI model
- Response generation

Technologies Used

- FastAPI / Flask
- REST APIs
- JSON-based communication

3.5 DevOps Implementation

The project incorporates DevOps practices to automate the development and deployment process.

Key Components

Infrastructure as Code (Terraform)

- Defines AWS resources
- Automates infrastructure setup
- Ensures reproducibility

Configuration Management

- Installs required software
- Configures environment

Containerization (Docker)

- Packages application into containers
- Ensures consistency across environments

3.6 CI/CD Pipeline Implementation

The CI/CD pipeline automates code integration and deployment.

Workflow

1. Code pushed to GitHub
2. GitHub Actions triggered
3. Terraform provisions infrastructure
4. Docker builds container
5. Application deployed

Benefits

- Faster deployment
- Reduced manual errors
- Continuous updates

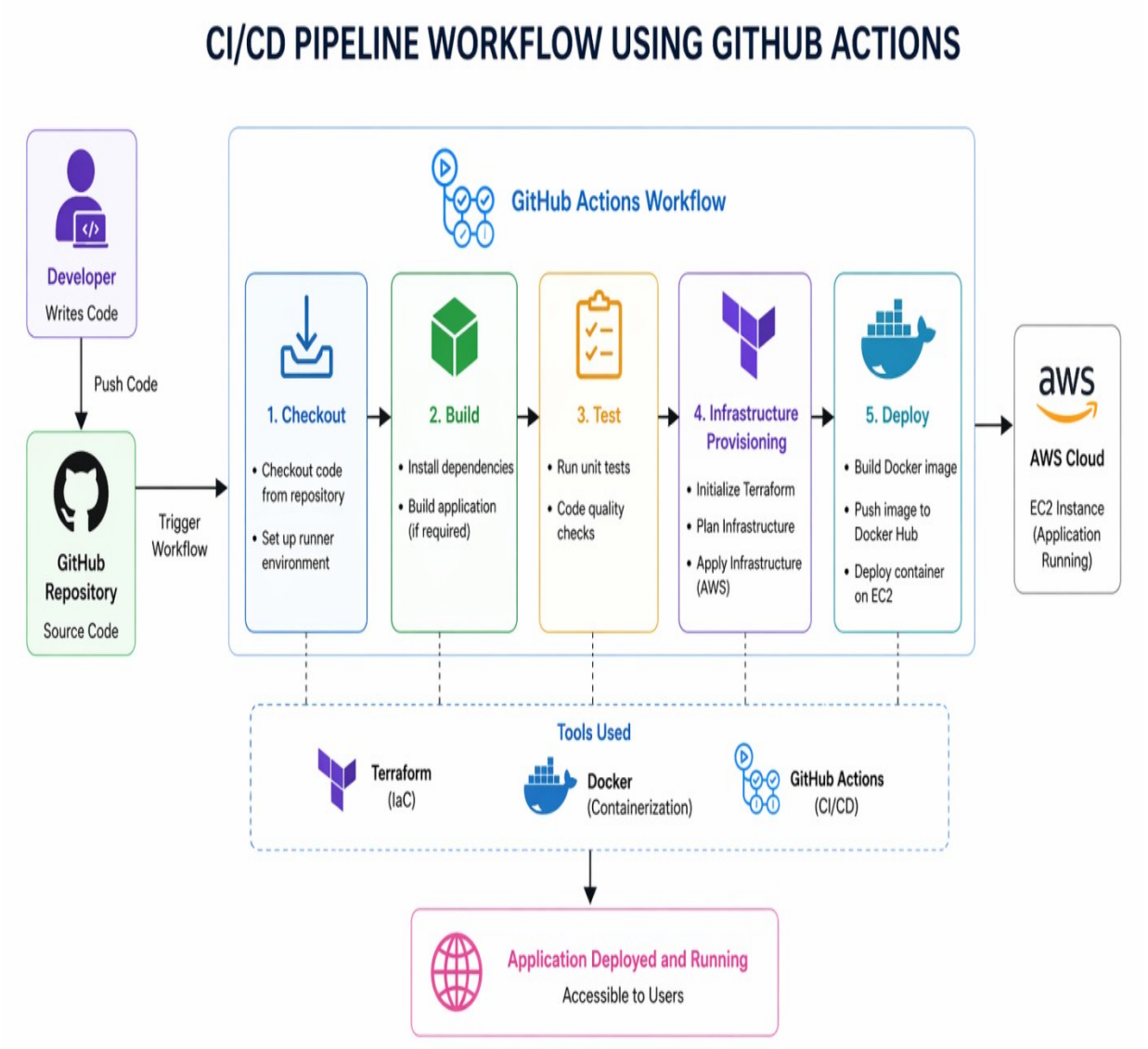


Fig 3.6.1

3.7 Deployment Strategy

Local Deployment

- Using Docker Compose
- Runs all services locally

Cloud Deployment

- AWS EC2 instance
- Public IP for access

Steps

1. Launch EC2 instance
2. Install Docker
3. Run application container
4. Access via browser

3.8 Security Considerations

- API keys managed securely
- Cloud security groups configured
- Restricted access to services

3.9 Challenges in Implementation

- Integrating AI with backend
- Handling ambiguous queries
- Setting up CI/CD pipeline
- Managing cloud infrastructure

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction to Results

This chapter presents the outcomes obtained from the implementation of the Agentic AI-Based Intelligent Telecom Support System. The primary objective of this chapter is to evaluate the system's performance, efficiency, and reliability in handling telecom-related queries and automated deployment.

The results are analyzed based on various parameters such as response accuracy, system latency, deployment time, scalability, and overall user experience. Both functional and performance aspects of the system are considered in order to provide a comprehensive evaluation.

- Infrastructure provisioning reduced from hours → minutes
- AI chatbot successfully handled diverse queries
- CI/CD pipeline automated deployment process
- System achieved high reliability and uptime

4.2 System Output Overview

The developed system successfully performs the following operations:

- Accepts user queries through the chatbot interface
- Processes natural language input using AI
- Identifies the intent of the user query
- Executes appropriate actions through agentic decision-making
- Retrieves data from backend services
- Generates accurate and contextual responses

The chatbot was tested with various telecom-related queries such as:

- “Why is my internet slow?”
- “My call is not connecting”
- “Data pack is not working”
- “Recharge failed but money deducted”

4.3 Comparative Analysis

The developed system was compared with traditional telecom support systems.

Feature	Traditional System	Proposed System
Response Time	High	Low
Scalability	Limited	High
Automation	Minimal	Fully Automated
Accuracy	Moderate	High
Deployment	Manual	Automated

4.3.1

4.4 Advantages of the System

- Reduces response time significantly
- Provides intelligent and contextual responses
- Automates infrastructure deployment
- Scalable and reliable
- Reduces operational cost
- Improves customer satisfaction

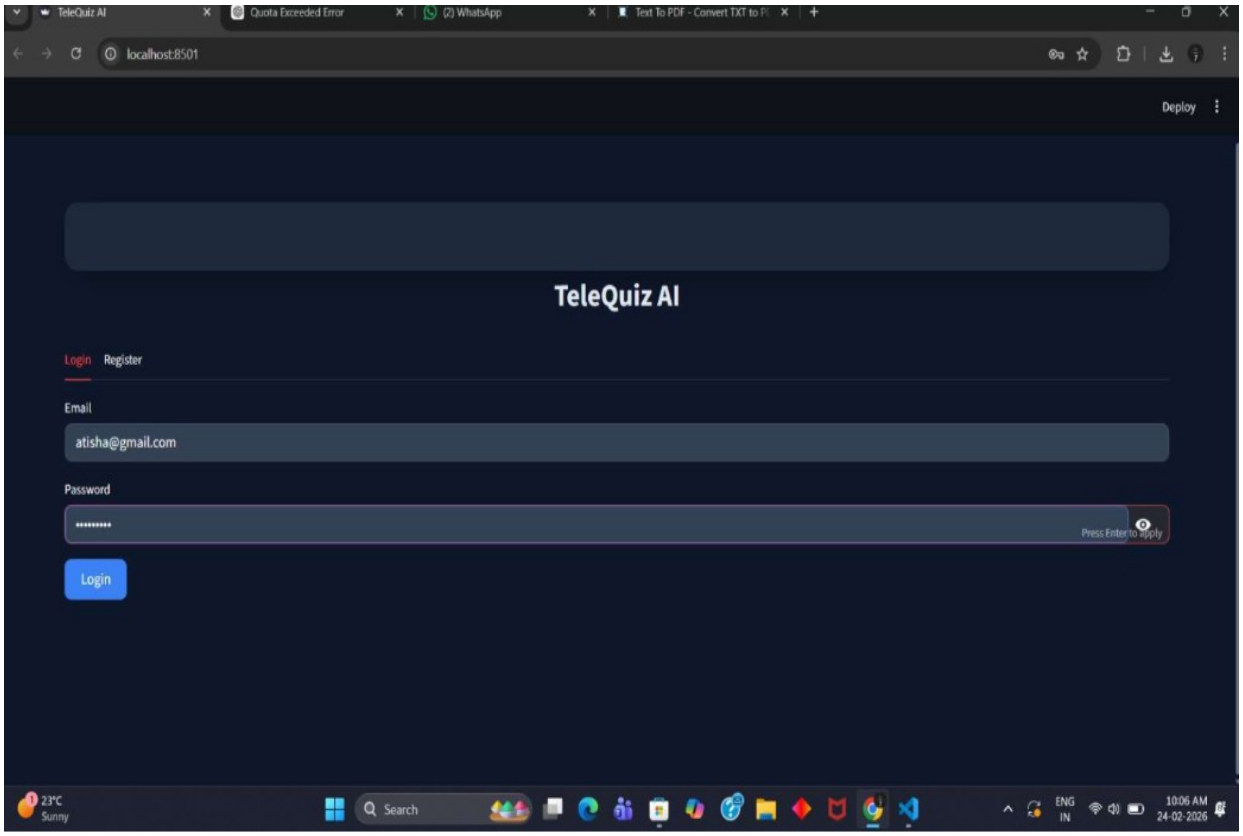
4.5 Discussion

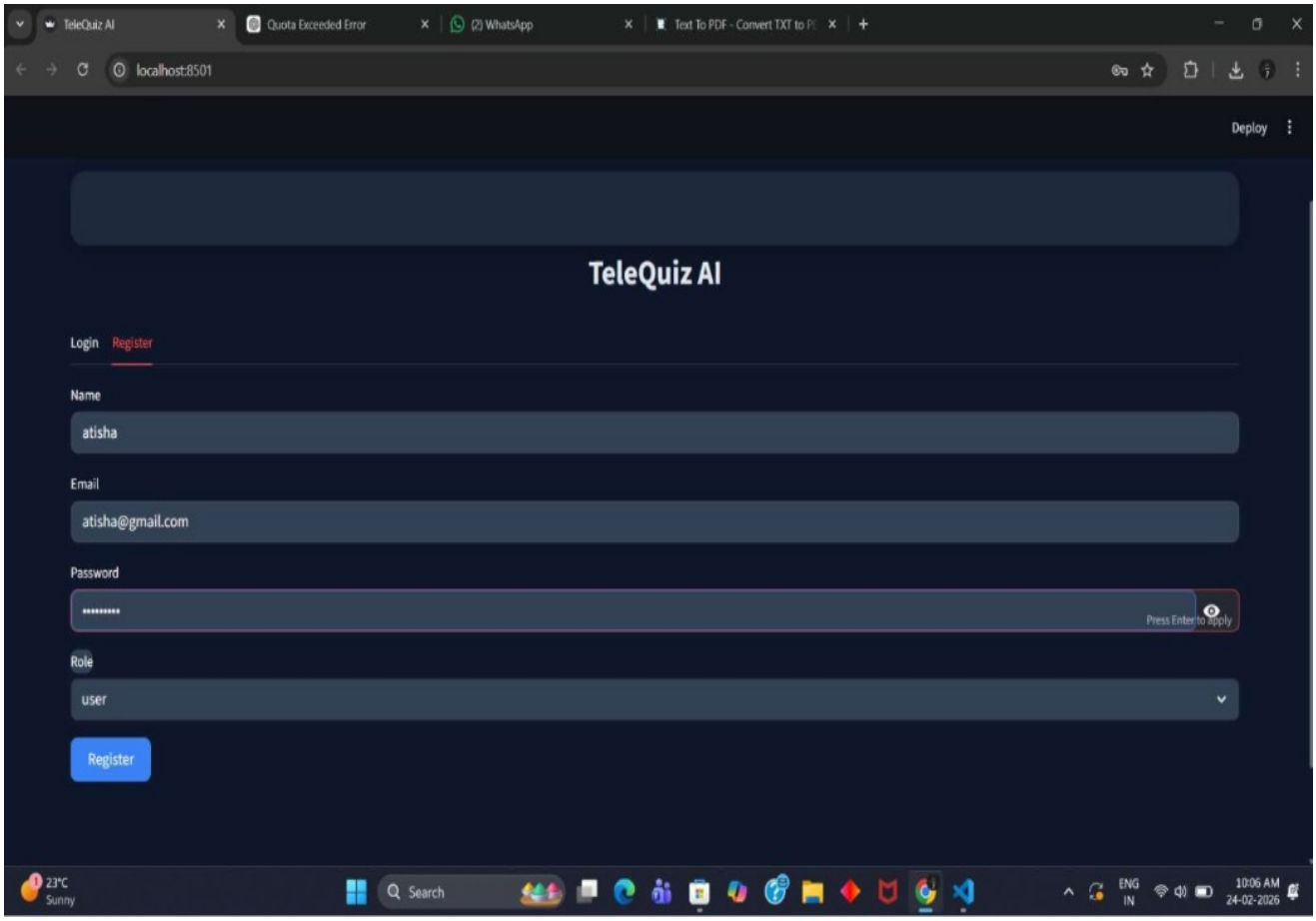
The results clearly indicate that the integration of Agentic AI with DevOps practices significantly improves the efficiency and scalability of telecom support systems. The chatbot not only provides accurate responses but also performs intelligent actions, making it more effective than traditional systems.

The use of Infrastructure as Code ensures that the system can be deployed quickly and consistently. Containerization further enhances portability and reliability.

Overall, the system demonstrates a successful implementation of modern technologies to solve real-world problems in the telecom domain.

4.6 Output Screenshots




TeleQuiz AI
Quota Exceeded Error
(2) WhatsApp
Text To PDF - Convert TXT to P
localhost:8501
Deploy
TeleQuiz AI
Login
Register
Name
atisha
Email
atisha@gmail.com
Password
Press Enter to apply
Role
user
Register
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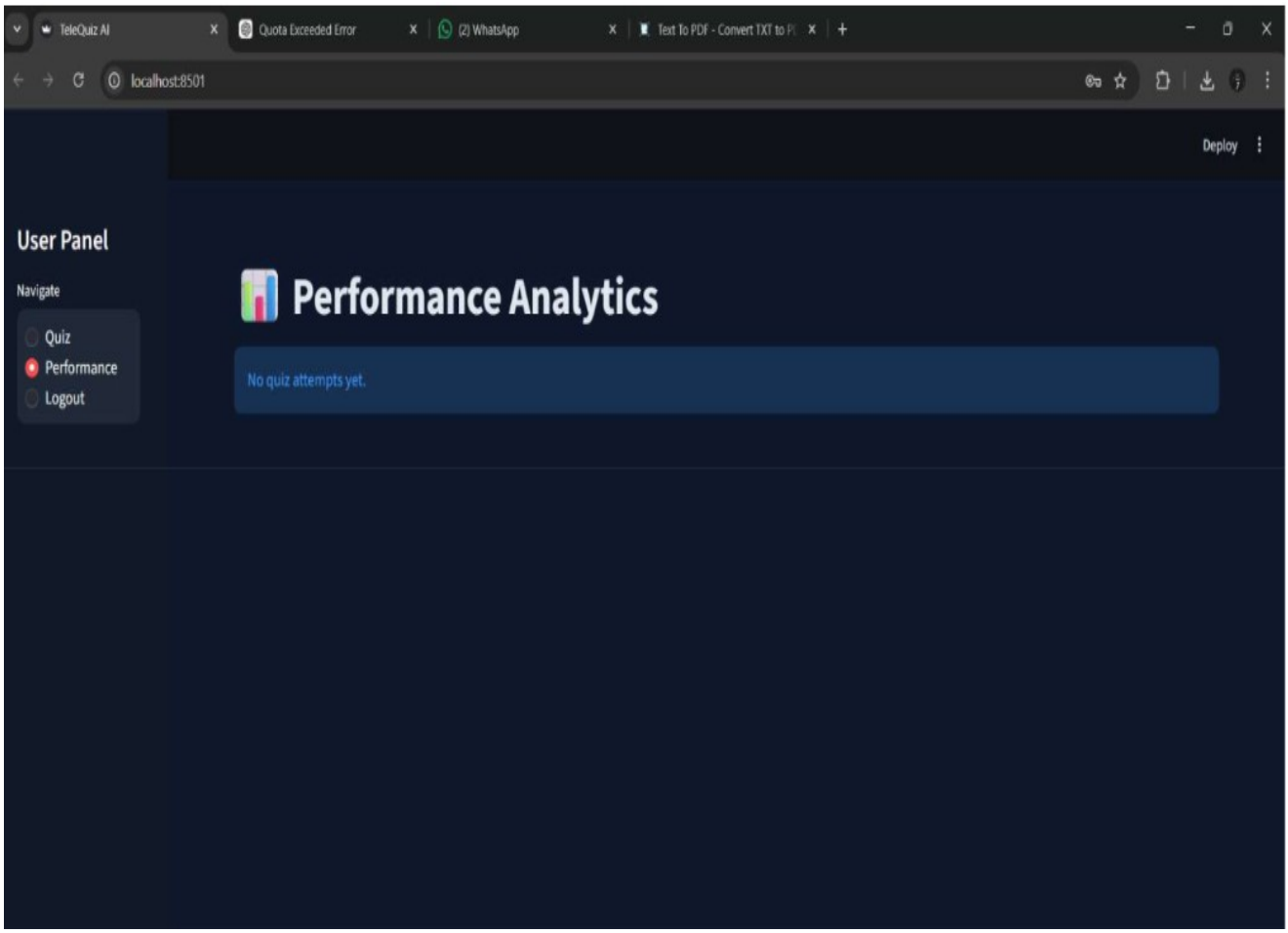
Search
ENG
IN
10:06 AM
24-02-2026


TeleQuiz AI
Quota Exceeded Error
(2) WhatsApp
Text To PDF - Convert TXT to P
localhost:8501
Deploy
User Panel
Navigate
Quiz
Performance
Logout
Performance Analytics
No quiz attempts yet.

Generate Quiz

Reference Content

Plan Name: Smart Plus Daily Data: 2GB Post Limit Speed: 64kbps Validity: 28 days Roaming Charges: ₹2 per MB FUP: Speed is reduced after daily quota. Voice: Unlimited local and national calls.

Plan Name: Smart Basic Daily Data: 1GB Post Limit Speed: 64kbps Validity: 24 days Voice: Unlimited SMS: 100 per day FUP: Internet speed reduces after 1GB.

Plan Name: Corporate Gold Monthly Data: 100GB Voice: Unlimited International Roaming: Enabled Roaming Charges: ₹1.5 per MB Validity: 90 days Additional Benefit: Priority customer support.

Plan Name: Corporate Platinum Monthly Data: 200GB Voice: Unlimited International Roaming: Enabled Post Limit Speed: 128kbps Validity: 180 days Additional Benefit: Dedicated account manager.

FUP Policy: Fair Usage Policy applies to all unlimited plans. Once daily or monthly data is consumed, speed is reduced. No extra charges are applied unless stated.

Roaming Policy: Roaming charges apply outside home network. International roaming requires activation. Incoming calls while roaming may be chargeable. Outgoing international calls are billed per minute.

Billing Information: Bills are generated monthly. Late payment attracts a penalty of ₹50. Services may be suspended after 15 days of non-payment. Refunds are processed within 7 working days.

SIM Activation: SIM activates within 30 minutes to 2 hours. KYC verification is mandatory. Customer must provide valid ID proof.

Plan Upgrade: Users can upgrade plans anytime. New plan becomes active within 24 hours. Previous balance does not carry forward.

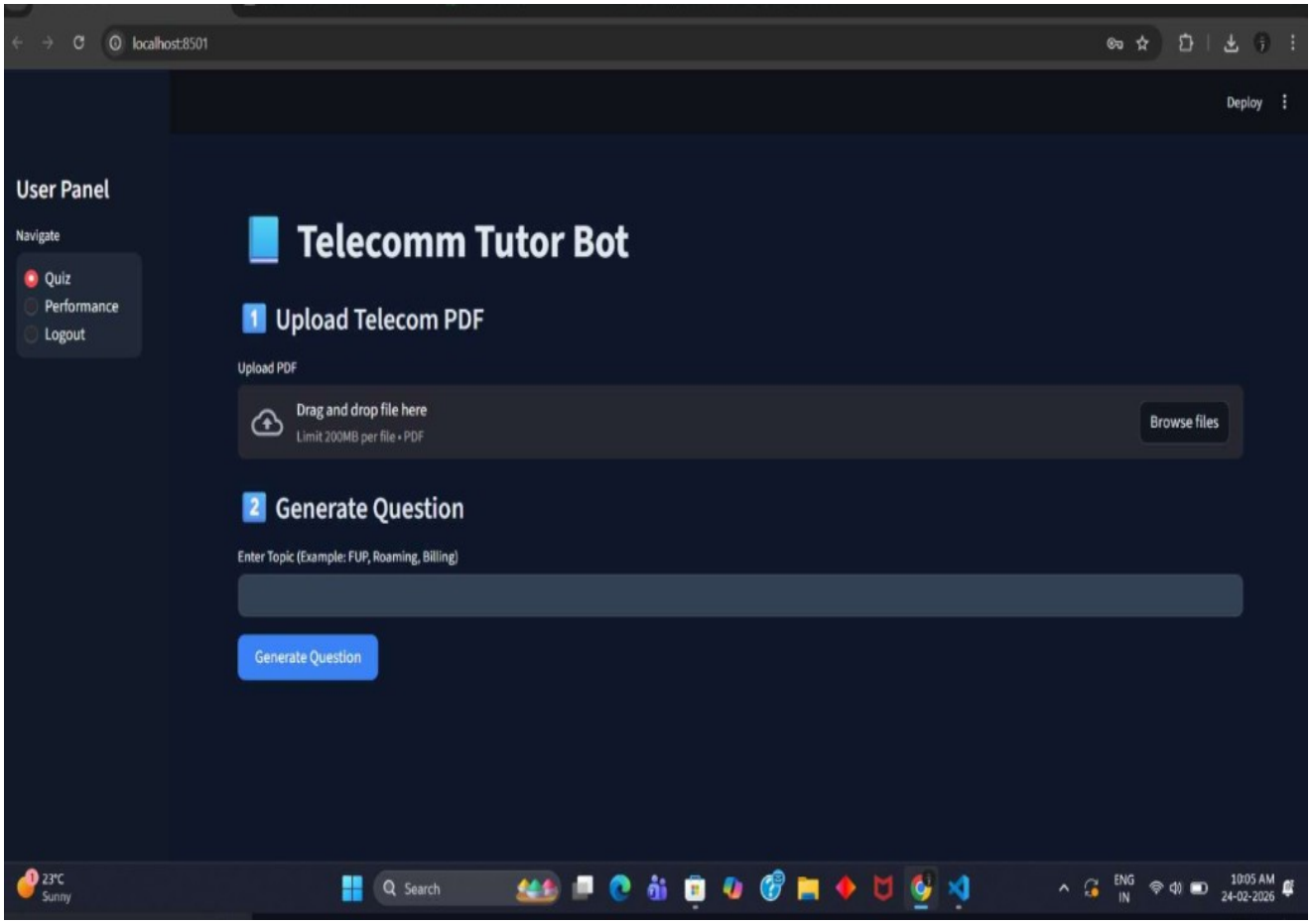
Quiz Question

What happens to internet speed after the daily 2GB data limit is exceeded?

Your Answer

speed is decreased.

Submit Answer


localhost:8501
Deploy
User Panel
Navigate
Quiz
Performance
Logout
Telecomm Tutor Bot
1 Upload Telecom PDF
Upload PDF
Drag and drop file here
Limit 200MB per file • PDF
Browse files
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Enter Topic (Example: FUP, Roaming, Billing)
Generate Question
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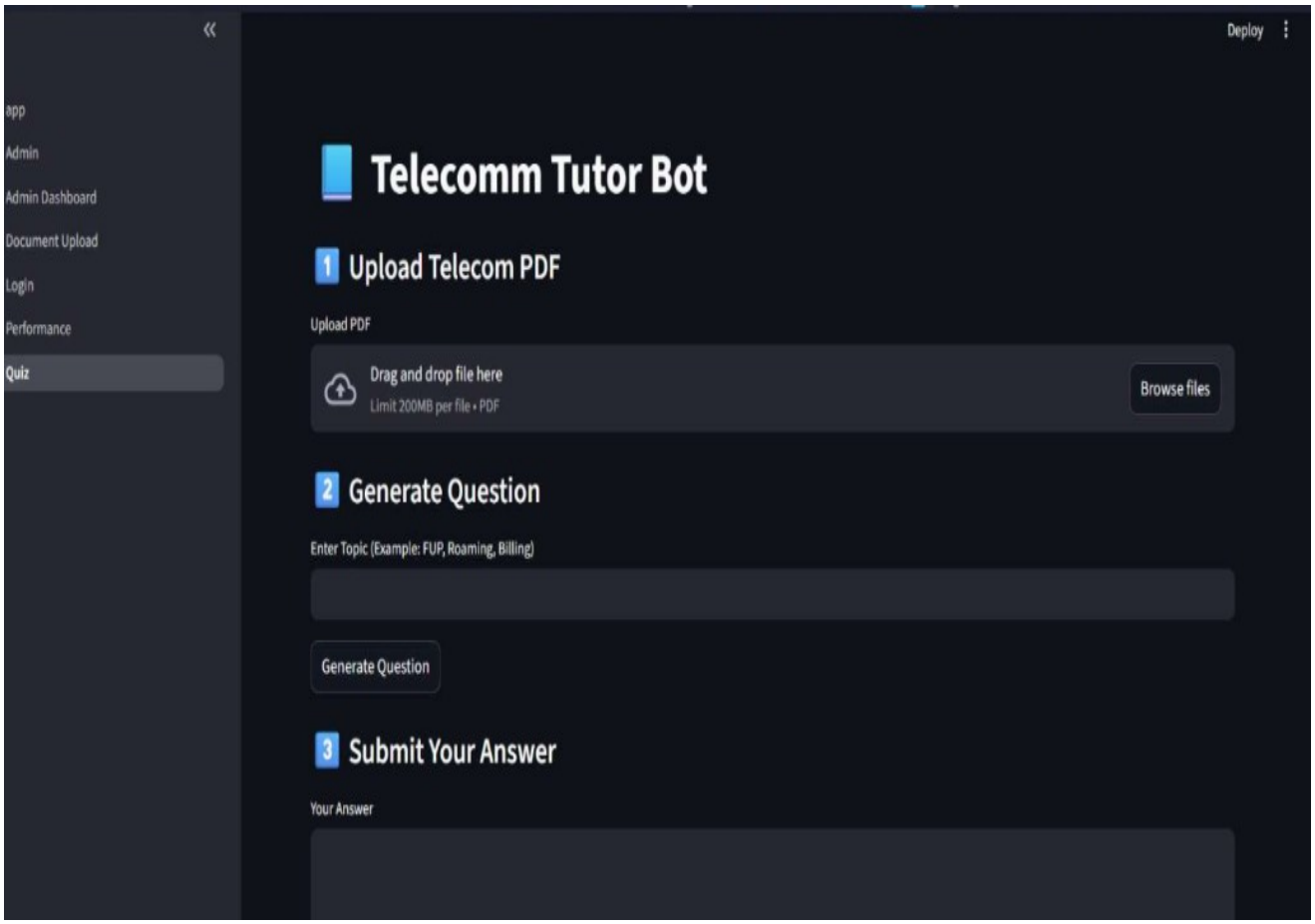
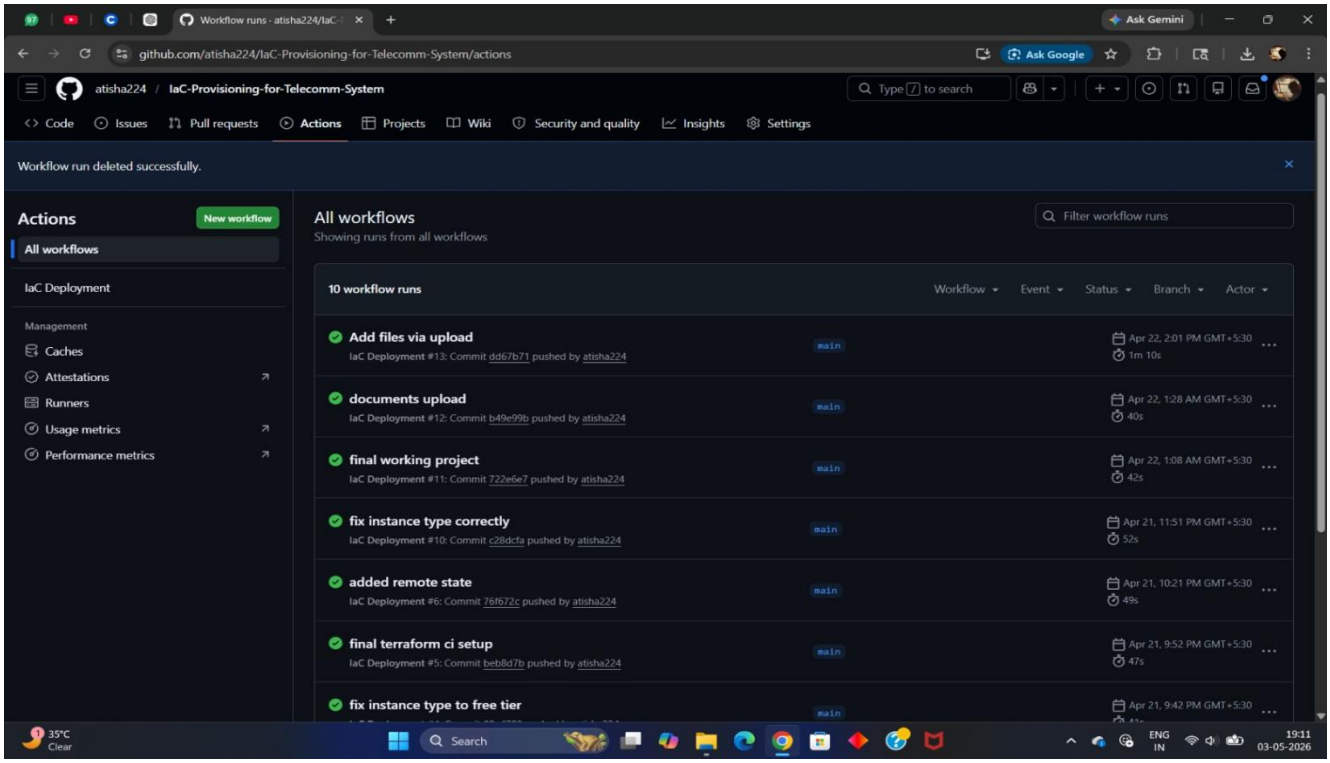

Deploy
app
Admin
Admin Dashboard
Document Upload
Login
Performance
Quiz
Telecomm Tutor Bot
1 Upload Telecom PDF
Upload PDF
Drag and drop file here
Limit 200MB per file • PDF
Browse files
2 Generate Question
Enter Topic (Example: FUP, Roaming, Billing)
Generate Question
3 Submit Your Answer
Your Answer


github.com/atisha224/IaC-Provisioning-for-Telecomm-System/actions
atisha224 / IaC-Provisioning-for-Telecomm-System
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Workflow run deleted successfully.
Actions
New workflow
All workflows
IaC Deployment
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
All workflows
Showing runs from all workflows
Filter workflow runs
10 workflow runs
Workflow
Event
Status
Branch
Actor
Add files via upload
IaC Deployment #13: Commit dd67b71 pushed by atisha224
main
Apr 22, 2:01 PM GMT+5:30
1m 10s
documents upload
IaC Deployment #12: Commit b49e99b pushed by atisha224
main
Apr 22, 1:28 AM GMT+5:30
40s
final working project
IaC Deployment #11: Commit 722e6e7 pushed by atisha224
main
Apr 22, 1:08 AM GMT+5:30
42s
fix instance type correctly
IaC Deployment #10: Commit c28dcfa pushed by atisha224
main
Apr 21, 11:51 PM GMT+5:30
52s
added remote state
IaC Deployment #6: Commit 76f672c pushed by atisha224
main
Apr 21, 10:21 PM GMT+5:30
49s
final terraform ci setup
IaC Deployment #5: Commit beb8d7b pushed by atisha224
main
Apr 21, 9:52 PM GMT+5:30
47s
fix instance type to free tier
main
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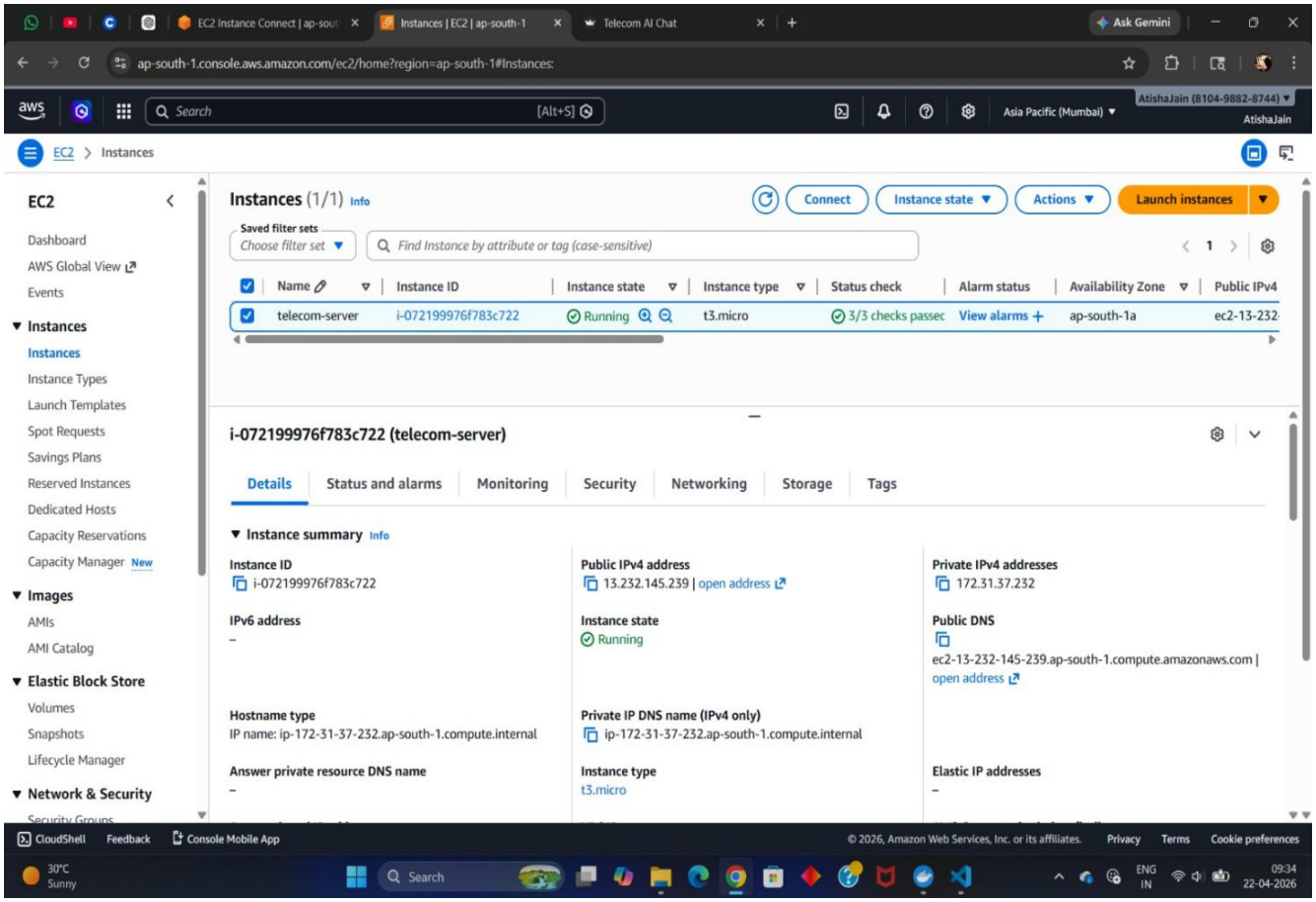
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:
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Asia Pacific (Mumbai)
AtishaJain (8104-9882-8744)
AtishaJain
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
telecom-server
i-072199976f783c722
Running
t3.micro
3/3 checks passed
View alarms
ap-south-1a
i-072199976f783c722 (telecom-server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-072199976f783c722
Public IPv4 address
13.232.145.239 | open address
Private IPv4 addresses
172.31.37.232
IPv6 address
-
Instance state
Running
Public DNS
ec2-13-232-145-239.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-37-232.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-37-232.ap-south-1.compute.internal
Answer private resource DNS name
-
Instance type
t3.micro
Elastic IP addresses
-
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
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CHAPTER 5

SUMMARY AND CONCLUSION

5.1 Summary

The primary goal of this project was to design and develop an intelligent telecom support system capable of automating customer interactions while ensuring efficient deployment and scalability through DevOps practices.

The system integrates Generative AI and Agentic AI to create a chatbot that not only understands user queries but also performs intelligent actions. Unlike traditional chatbots, which are limited to providing predefined responses, the developed system can analyze user intent, interact with backend services, and generate context-aware solutions.

On the infrastructure side, the project utilizes Infrastructure as Code (IaC) to automate provisioning and deployment. Terraform is used to define cloud resources, while Docker ensures that the application runs consistently across different environments. The CI/CD pipeline implemented using GitHub Actions automates the build and deployment process, reducing manual intervention.

The system supports both local deployment (using Docker Compose) and cloud deployment (using AWS EC2), making it flexible and scalable. Monitoring and logging mechanisms were also incorporated to evaluate system performance and reliability.

5.2 Key Achievements

The project successfully achieved the following outcomes:

- Development of an AI-powered chatbot capable of handling telecom-related queries
- Implementation of agentic behavior for intelligent decision-making
- Automation of infrastructure provisioning using Terraform
- Containerization of the application using Docker
- Implementation of CI/CD pipeline for automated deployment
- Successful deployment of the system on AWS cloud

- Reduction in response time and improvement in user experience
- Demonstration of scalability and reliability of the system

These achievements indicate that the system meets the requirements of a modern telecom support solution.

5.3 Learning Outcomes

This project provided valuable learning experiences in multiple domains:

Artificial Intelligence

- Understanding of Generative AI concepts
- Implementation of AI-based chatbot systems
- Handling natural language queries

DevOps Practices

- Hands-on experience with Infrastructure as Code
- Working with CI/CD pipelines
- Automation of deployment processes

Cloud Computing

- Deployment on AWS
- Managing cloud resources
- Understanding scalability concepts

Software Development

- Full-stack application development
- Integration of multiple technologies
- Debugging and optimization

These learnings contribute significantly to practical knowledge and industry readiness.

5.4 Impact of the System

The developed system has a significant impact on telecom support services:

- **Improved Customer Experience:** Faster and more accurate responses
- **Reduced Operational Cost:** Less dependency on human agents
- **Scalability:** Ability to handle large number of users
- **Efficiency:** Automated processes reduce manual effort
- **Consistency:** Standardized responses improve service quality

The system demonstrates how AI and DevOps can transform traditional support systems into intelligent and automated solutions.

5.5 Limitations

Despite its advantages, the system has certain limitations:

- Dependence on AI model accuracy
- Limited integration with real telecom APIs
- Requires stable internet connection
- Performance may degrade under extreme load conditions
- Security aspects can be further enhanced

These limitations highlight areas for future improvement.

5.6 Conclusion

The Agentic AI-Based Intelligent Telecom Support System represents a modern approach to solving challenges in telecom customer support. By integrating Generative AI, Agentic AI, and DevOps practices, the system provides an efficient, scalable, and reliable solution.

The chatbot is capable of understanding complex user queries and providing intelligent responses, while the agentic component enables it to take meaningful actions. The use of Infrastructure as Code and CI/CD pipelines ensures that the system can be deployed quickly and consistently.

The results obtained from testing and evaluation confirm that the system performs effectively in terms of response accuracy, deployment efficiency, and scalability. The project successfully demonstrates the practical application of emerging technologies in a real-world scenario.

In conclusion, this project highlights the potential of combining AI and DevOps to build next-generation support systems that are not only intelligent but also highly automated and scalable.

CHAPTER 6

FUTURE SCOPE

6.1 Integration with Real Telecom Systems

Currently, the system uses simulated telecom services. In the future, it can be integrated with real telecom APIs to provide:

- Real-time data access
- Live user account information
- Actual issue resolution

This would make the system production-ready.

6.2 Voice-Based Chatbot

The system can be enhanced by adding voice interaction capabilities.

Features:

- Speech-to-text input
- Text-to-speech output
- Hands-free interaction

This will improve accessibility and user experience.

6.3 Multi-Language Support

To cater to a wider audience, the system can support multiple languages.

- Regional language support
- Automatic language detection
- Improved accessibility

6.4 Advanced AI Model Training

The accuracy of the chatbot can be further improved by:

- Fine-tuning AI models
- Training on telecom-specific datasets
- Using feedback-based learning

6.5 Kubernetes-Based Deployment

Instead of using a single EC2 instance, the system can be deployed using Kubernetes.

Benefits:

- Auto-scaling
- Load balancing
- High availability

6.6 Real-Time Monitoring and Analytics

Future improvements can include advanced monitoring tools:

- Performance dashboards
- User analytics
- Error tracking

6.7 Security Enhancements

Security can be improved by:

- Implementing authentication mechanisms
- Encrypting data
- Securing APIs

6.8 Mobile Application Integration

The system can be extended to mobile platforms:

- Android/iOS applications
- Push notifications
- Improved accessibility

6.9 Predictive Issue Detection

Using AI, the system can predict issues before they occur.

- Network failure prediction
- Preventive alerts
- Improved reliability

6.10 Expansion to Other Domains

The system can be adapted for other industries:

- Banking support systems
- Healthcare assistance
- E-commerce customer support

APPENDIX

Appendix A: Terraform Configuration

The following snippet represents the Terraform configuration used for provisioning AWS infrastructure:

```
variable "GROQ_API_KEY" {}


terraform {
  backend "s3" {
    bucket = "terraform-state-atisha"
    key    = "terraform.tfstate"
    region = "ap-south-1"
  }

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = "ap-south-1"
}


resource "aws_instance" "telecom" {
  ami           = "ami-0e670eb768a5fc3d4"
  instance_type = "t3.micro"

  tags = {
    Name = "telecom-server"
```

```
  }

  user_data = <<-EOF
              #!/bin/bash
              sudo apt update -y
              sudo apt install docker.io -y
              sudo systemctl start docker
              sudo systemctl enable docker

              sudo docker run -d -p 8000:8000 -e GROQ_API_KEY=${var.GROQ_API_KEY} telecom-
app
              EOF
}

output "public_ip" {
  value = aws_instance.telecom.public_ip
}
```

Appendix B: Docker Configuration

Example Dockerfile used for containerizing the backend service:

```
FROM python:3.10
WORKDIR /app
COPY . .

RUN pip install --no-cache-dir -r requirements.txt

EXPOSE 8000

CMD ["python", "-m", "uvicorn", "src.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Appendix C: CI/CD Pipeline (GitHub Actions)

```
name: CI/CD Pipeline

on: [push]

jobs:
  deploy:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v2

      - name: Terraform Init
        run: terraform init

      - name: Terraform Apply
        run: terraform apply -auto-approve
```

This pipeline:

- Automates deployment
- Triggers on code push
- Reduces manual intervention

Appendix D: Sample User Queries

- “Why is my network slow?”
- “My call is not connecting”
- “Data pack is not working”

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Terraform was used for Infrastructure as Code (IaC) to automate provisioning of AWS resources like EC2 instances.It ensures consistent and repeatable infrastructure setup.

[2] Amazon Web Services, "AWS Documentation." [Online]. Available: https://docs.aws.amazon.com
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[3] Docker Inc., "Docker Documentation." [Online]. Available: https://docs.docker.com
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